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(1) Research Program/Project/Study Title: Biosecurity and ASF Preparedness of Pig Raisers Against Potential African Swine Fever outbreak in the Province of Capiz	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☒ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center							
(4) Site of implementation									
Municipality/City:									
Province:		Capiz							
Region:		VI							
(5) Classification <table border="0"> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☒ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☐ Applied</td> <td>☐ Tech.</td> </tr> </table> Promotion/Commercialization		Research:	Development:	☒ Basic	☐ Pilot Testing	☐ Applied	☐ Tech.	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
Research:	Development:								
☒ Basic	☐ Pilot Testing								
☐ Applied	☐ Tech.								
(7) Sector/Commodity Livestock		(8) Discipline/Field of Specialization Swine							

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(9) Rationale

African Swine Fever (ASF) is a highly contagious and fatal viral disease affecting young and adult pigs however it does not pose risk to human health. Accordingly, the Philippines has been the 9th country in Asia to have the case. The first outbreak in the Philippines was recorded in July 2019 and as of July 2022 (DA) there were 53 provinces, 704 municipalities, and 3832 barangays have experienced an ASF outbreak. In Western Visayas, the first reported and confirmed case was on Oct 13, 2022, from Iloilo Province. According to reports, 23 out of 73 barangays in Oton have positive cases of ASF in addition, an outbreak in San Miguel was also confirmed. The unavailability of a vaccine against a complex ASF DNA virus pose a great challenge for a comprehensive approach to contain and control its impact.

The Province of Capiz where the study will be conducted is still free from ASF but the risk is high considering the strategic location as well as the volume of pigs and pig products coming from neighboring provinces including Iloilo. The province of Capiz has already issued an EO number 24 mandating the provincial task force to intensify border controls, conduct checkpoints, and implement strict veterinary quarantine procedures and food safety measures on the movement of swine, pork, and pork products and by-product in the province of Capiz.

Preventing the further spread of ASF from the initially affected areas appears to be the core strategy to contain the economic losses caused by ASF (Gain, 2019). The Food and Agriculture Organization (FAO) through the Agriculture and Consumer Protection Department emphasizes early reaction, detection, and notification including the application of strict biosecurity measures. This also includes improved husbandry practices, disinfection, and good surveillance and monitoring of live pigs being transported (FAO, 2020). Stringent compliance with biosecurity measures and cooperation with government initiatives are key strategies to prevent the introduction and contain the impact of ASF.

The aim of this study is to determine and understand the current ASF awareness and document the biosecurity practices of backyard pig farmers within the province of Capiz, Philippines.

The results of this study will provide information to policymakers government units and pig farm owners of both backyard and commercial the threat posed by possible ASF outbreak in the province of Capiz.

(9-A) Project Description

(10) Objectives of the Study

The study aims to determine the biosecurity and preparedness of in the different municipalities of the Province of Capiz

Specifically, aims to answer the following:

1. What is the socio-economic and demographic profile of pig farmers?
2. What is the level of knowledge of pig farmers about ASF?
3. What is the production status of the piggery farm?
4. What are the management and biosecurity practices of the pig farm?

(11) Review of Literature/Prior Art Search

African Swine Fever

African swine fever is an acute hemorrhagic disease of pigs, a re-emerging viral disease that threatening the global pig industry (Sánchez-Cordón, 2018). The virus is a large DNA virus and is the only member of the *Asfarviridae* family. In pigs, the virus replicates predominantly in macrophages. ASFV is a large double-stranded DNA virus and is the only member of the *Asfarviridae* family (Alonso, 2018). The genome varies in length between approximately 170 and 193 kbp, mainly owing to gain or loss of multigene

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family members. Many ASFV genes inhibit host defenses, including type I interferon, the main host antiviral pathway, and programmed cell death, or apoptosis

Status of ASF in Asia and Philippines

Update of the FAO (2022), ASF outbreaks has been reported in countries in Asia such as China (2018), Mongolia (2019), Viet Nam (2019), Cambodia (2019), Korea (2019), Lao (2019), Myanmar (2019), Timor-Leste (2019), Indonesia (2019), Papua New Guinea (2020), India (2020), Malaysia(2021), Bhutan (2021), Thailand (2022) and Nepal (2022). The first outbreak of ASF in the Philippines started in July 2019 and as of July 2022, there are approximately 53 provinces, 704 cities/ municipalities and 3,832 barangays have cases of ASF outbreaks (<https://www.pna.gov.ph/articles/1178465>). Backyard hog farmers in the Province of Cebu are advised to observe biosecurity measures in their farms to prevent the spread of diseases affecting pigs amid the recent African swine fever (ASF) outbreak in different places in Luzon in 2019. (<https://www.pna.gov.ph/articles/1081296>). The disease has also been reported in 49 towns and cities in Eastern Visayas (<https://www.pna.gov.ph/articles/1154395>) and has affected 6.68% of the region's total hog population.

Iloilo and the whole Western Visayas region had previously been among the last few areas in the country that had remained ASF-free since the first reported outbreak in July 2019 (<https://ptvnews.ph/da-western-visayas-told-to-test-pig-breeding-sites-to-curb-spread-of-asf/>). On 13 October 2022, the first ASF cases in Western Visayas Region were reported from Iloilo Province (<https://www.pna.gov.ph/articles/1186607>). In Oton, 23 out of its 37 barangays have positive cases (<https://www.pna.gov.ph/articles/1186993>); an outbreak was also confirmed in neighboring San Miguel Municipality, Iloilo Province (<https://www.pna.gov.ph/articles/1186993>).

As to date the Republic of the Philippines have been taking actions these include the reduction of tariff rates of imported pork was extended until December 31, 2022 (<https://www.officialgazette.gov.ph>). Further, local authorities are mandated to strictly follow the National Zoning implementation and movement plan depending on the level of ASF risks. In the Visayas region, municipalities of Oton and San Miguel were considered as red zone; the Iloilo Province issued the movement ban of pigs and pork products from the red zone for at least 10 days. Culling operations were conducted within 0.5 km radius of the affected areas in Iloilo Province instead of 1 km in the previous implementation as stipulated in a new DA memorandum.

Other provinces in Western Visayas namely Antique, Negros Occidental, Cebu, Aklan, Guimaras, and Capiz have strengthened its border control to prevent the entry of pigs and pork products from ASF affected areas, particularly from Iloilo Province. A joint task force against ASF has also been established to implement preventive measures in Negros Occidental Province. The President of the Philippines ordered Department of Agriculture Regional Office 6 (DA-6) to test all pig breeding sites in the Western Visayas Region to stop the spread of the disease. DA-6 also called on the assistance of stakeholders to protect "green zones" in Iloilo Province to stop the spread of ASF.

Commercial and Backyard pig raisers

Backyard hog farmers/ farming (<https://www.da.gov.ph>) is defined as the household or farm-based hog raiser/ raising with 1-20 heads of growing or adult pigs (excluding sucklings) in the vicinity of their residences and properties while a piggery farm with a capacity of at least 21 head adult and 22 heads young is considered a commercial (<https://psa.gov.ph>). Both commercial and backyard production, farm biosecurity is crucial to avoid ASF introduction and spread on pig farms. However, backyard farms represent the weakest part of pig production and so, most outbreaks have been notified on such farms (<https://www.pig333.com>). Measures to mitigate the risk of the spread of ASFV in different commercial pig production systems and backyard holdings have been analyzed by Bellini et al. (2016).

Biosecurity measures

Biosecurity was defined as "the security form of transmission of infectious diseases, parasites and pests" (Amas,1990). Biosecurity involves a set of management and physical measures designed to reduce the risk of introduction, establishment and spread of animal diseases, infections, or infestations to, from, and

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within an animal population (glossary TAHC OIE, 2018). The elements necessary for the implementation of biosecurity include the farm, the location, and the epidemic situation. Controlling ASF has proven to be a challenge because no vaccine is available, the disease has a very high socioeconomic impact and threatens the livelihoods of resource-poor farmers (Costar 2009) and the only control measures available are strict quarantine, and biosecurity, animal movement restrictions and slaughtering affected/exposed animals (Sánchez-Cordón, 2018).

Epidemiology

ASF is mainly transmitted by direct contact between infected and susceptible pigs or through ingestion of the virus contaminated pork products (Costard et al., 2013; Guinat et al, 2016). The virus is extremely resistant to sunlight and putrefaction and can persist in refrigerated meat and carcasses for 6 months and longer when frozen.

ASF Awareness

Pig farmers having poor knowledge about ASF clinical signs and limited concern about ASF are less likely to consider the possibility of an outbreak in their farm as cited by Guinat et al., (2016). Moreover, pig farmers lacking awareness of outbreaks in other countries, have a perception of the negative impact on them resulting from false positive reporting and the perceived complexity of reporting processes as less likely to report ASF suspicion. Their findings indicate important areas for educational campaigns to pig farmers to focus on the attempt to increase the rapid response to ASF outbreak. Increased knowledge of the routine management and biosecurity practices of pig farmers and the key herd health programs likely to be associated with those practices would be of value as the basis of a risk-based approach to disease surveillance and control (Alawneha et al, 2014).

(12) Scientific Basis/Theoretical Framework

The study of Bernardes and Peña showed that pig farmers were generally aware of the threat posed by the African Swine Fever disease and have relatively good biosecurity practices, only 32.90% were convinced that the Baybay City in Leyte is prepared in the case of an ASF outbreak. Vaccination, footbath, and control swill feeding need to be emphasized. Further suggestion on the additional effort on the preparation and training of pig farmers on disease monitoring and surveillance and improving biosecurity practices.

(13) Methodology

Date and Place of Research

The research will be conducted from December 2022 to July 2023 in selected barangays in the municipalities of Capiz with the highest number of farms. Data on the piggery farms will be taken from the respective local municipal government.

The farmer respondents will be randomly selected (snowball/ purposive sampling) and proportionally allocated per municipality and barangay from a given population of backyard pig raisers provided by the local agriculture office.

Materials

The research will use a structured questionnaire and conduct a scheduled interview. The questionnaire will be constructed in English and translated into the local dialect (Hiligaynon/ilonggo) for ease and convenience during the one-on-one interview at the respective residence of the farmer respondents with prior verbal consent.

Any information that has the potential to identify the farmer-respondent will be handled with confidentiality and will not be included in the analysis to protect the privacy of respondents. In the event the first respondent will not be available or will refuse to participate, the next available backyard pig raiser closest to that respondent will be interviewed instead. The actual interview time per respondent will also be recorded.

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Research Design

The study will be using descriptive statistics for farm characteristics and pig farmer's awareness related to ASF.

Data Analysis

Data gathered will be analyzed using Microsoft Excel 2016 data analysis tool.

(14) Major Activities/Work plan

No	Description of activities	Times (Month)							
		December 2022	January 2023	February 2023	March 2023	April 2023	May 2023	June 2023	July 2023
1	Administrative preparation and material tools and permits								
2	Procurement of research materials								
3	Research implementation								
4	Preparation of reports								
5	Seminar / publication								

(15) Expected Output

- ✓ Provide information and document on the biosecurity measures practiced by piggery farms
- ✓ Increased information on the awareness, knowledge and management practiced by backyard raisers
- ✓ IEC materials on ASF for pig farmers
- ✓ Research paper for scientific conference and journal publication

(15) Technology Transfer Activity Design

The research will help backyard piggery farmers and technical personnel with the biosecurity measures, and knowledge needed for the preparation in case of African Swine Fever (ASF) outbreak in the Province of Capiz

(16) Target Beneficiaries

- ✓ Pig farmers
- ✓ Pork consumers and processors
- ✓ LGU livestock technicians, researchers and veterinarians
- ✓ Policy makers

(17) Personnel Requirement

Leah A. Ingalla	Member	Veterinary Medicine/Animal Science
Maryneth B. Barrios, DVM, MS	Member	Veterinary Medicine/Animal Science

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

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Program/Project Members (in hours):

(18) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)	1 @ 11,068 x 6 months	66,408
Transportation allowance		30,000
Professional fees (enumerators)		35,000
Supplies		20,000
Communication		10,000
Others		30,000
Total		191,408

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(20) Capsule Curriculum Vitae (One-page brief C.V. only)